LABORATORY OBSERVATION

Course: Full Stack Web Development

Course Code: 23CM4121

Experiment No.: 2

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1. TITLE

Design of a Styled Webpage with CSS3 Flexbox Layout and HTML5 Canvas Graphics

2. AIM

To develop a modern styled webpage applying external CSS3 gradients, navigation bar hover effects, flexbox cards, and dynamically drawing 2D shapes on an HTML5 canvas using JavaScript.

3. OBJECTIVE

The objectives of this experiment are:

- To master external CSS3 stylesheet linking and global styling.
  \n
- To apply linear gradients and text shadows to header elements.
  \n
- To implement responsive flexbox layouts using display: flex and gap.
  \n
- To understand the HTML5 <canvas> coordinate system and 2D rendering context.
  \n
- To draw rectangles, arcs, lines, and text programmatically via JavaScript.

4. THEORY

CSS3 introduces advanced visual styling capabilities including multi-color linear gradients, box shadows, and Flexbox for effortless alignment and distribution of UI cards. The HTML5 <canvas> element provides a scriptable 2D drawing surface. Using JavaScript's getContext('2d'), developers can programmatically render vector shapes, paths, strokes, fills, and custom text onto the canvas.

5. ALGORITHM / PROCEDURE

1. Create index.html and link external stylesheet style.css.
   \n
2. Design a gradient header using linear-gradient().
   \n
3. Build a centered navigation bar with link hover color transitions.
   \n

4. Declare an HTML5 canvas element with id='myCanvas'.
   \n
5. Implement a .container with display: flex and gap containing styled cards.
   \n
6. Write JavaScript to get the canvas 2D context using getContext('2d').
   \n
7. Use fillRect() to draw a rectangle, arc() to draw a circle, moveTo()/lineTo() for a line.
   \n
8. Test and verify the styled webpage and canvas graphics in a browser.

6. CODE EXPLANATION

The webpage is styled using an external CSS stylesheet linked in the HTML head. A smooth color transition is applied to the header background using linear-gradient(...). The layout of the child card elements is handled gracefully using Flexbox (display: flex) with a gap spacing of 30px. In the JavaScript section, canvas.getContext('2d') is called to acquire the 2D rendering context, which is then used to programmatically draw a rectangle using fillRect(...) and a circular path using arc(...).

7. PROGRAM

index.html

```
<!DOCTYPE html>
<html lang="en">

<head>

    <meta charset="UTF-8">

    <meta name="viewport" content="width=device-width, initial-scale=1.0">

    <title>My Styled Webpage</title>

    <!-- External CSS -->
    <link rel="stylesheet" href="style.css">

    <!-- Google Font -->
    <link href="https://fonts.googleapis.com/css2?
family=Poppins:wght@300;500;700&display=swap" rel="stylesheet">

</head>

<body>

    <!-- ================= HEADER ================= -->

    <header>

        <h1>ANITS Web Technology Showcase</h1>

        <p>
            HTML5 Graphics & CSS3 Modern Web Styling
        </p>

    </header>


    <!-- ================= NAVIGATION ================= -->

    <nav>

        <a href="#">🏠 Home</a>

        <a href="#canvas">🎨 Canvas</a>

        <a href="#cards">💻 Technologies</a>

        <a href="#footer">📞 Contact</a>

    </nav>


    <!-- ================= HERO SECTION ================= -->

    <section class="hero">

        <div class="hero-text">

            <h2>Welcome to My Styled Webpage</h2>
```

```
        <p>

            This webpage demonstrates HTML5 Canvas and CSS3 styling
            techniques including gradients, shadows,
            hover effects and modern layouts.

        </p>

        <button>Explore More</button>

    </div>

</section>


<!-- ================= CANVAS ================= -->

<section id="canvas">

    <h2 class="section-title">

        HTML5 Canvas Graphics

    </h2>

    <canvas id="myCanvas" width="850" height="350">

        Your browser does not support Canvas.

    </canvas>

</section>


<!-- ================= CARDS ================= -->

<section id="cards">

    <h2 class="section-title">

        Web Technologies

    </h2>

    <div class="card-container">

        <!-- Card 1 -->

        <div class="card">

            <h2>💻 HTML5</h2>

            <p>

                HTML5 provides semantic elements,
                multimedia support,
                forms,
                canvas
```

```
                and responsive webpage structure.

            </p>

            <button>Learn HTML5</button>

        </div>

        <!-- Card 2 -->

        <div class="card">

            <h2>🎨 CSS3</h2>

            <p>

                CSS3 enhances webpages using gradients,
                shadows,
                transitions,
                animations,
                hover effects
                and modern layouts.

            </p>

            <button>Learn CSS3</button>

        </div>

        <!-- Card 3 -->

        <div class="card">

            <h2>🖌 Canvas</h2>

            <p>

                Canvas allows drawing
                rectangles,
                circles,
                lines,
                text,
                games,
                animations
                and graphics using JavaScript.

            </p>

            <button>Learn Canvas</button>

        </div>

    </div>

</section>


<!-- ================= FOOTER ================= -->
```

```
<footer id="footer">

    <h3>

        Developed By

    </h3>

    <p>

        Karimajji Vinay

    </p>

    <p>

        CSE (AI & ML)

    </p>

    <p>

        Anil Neerukonda Institute of Technology and Sciences (ANITS)

    </p>

    <br>

    <p>

        © 2026 All Rights Reserved

    </p>

</footer>

<!-- ================= CANVAS SCRIPT ================= -->

<script>

    const canvas = document.getElementById("myCanvas");

    const ctx = canvas.getContext("2d");

    /* Rectangle */

    ctx.fillStyle = "#2563eb";

    ctx.fillRect(40, 40, 180, 120);

    /* Circle */

    ctx.beginPath();
```

```
        ctx.arc(380, 100, 60, 0, Math.PI * 2);

        ctx.fillStyle = "#10b981";

        ctx.fill();


        /* Line */

        ctx.beginPath();

        ctx.moveTo(520, 40);

        ctx.lineTo(780, 160);

        ctx.strokeStyle = "#ef4444";

        ctx.lineWidth = 5;

        ctx.stroke();


        /* Text */

        ctx.font = "30px Poppins";

        ctx.fillStyle = "#111827";

        ctx.fillText("Karimajji Vinay", 240, 280);

    </script>

</body>

</html>
```

style.css

```
*{
margin:0;
padding:0;
box-sizing:border-box;
}

body{
font-family:'Poppins',sans-serif;
background:linear-gradient(135deg,#8fb1de,#c9d2db,#56d6e2);
color:#333;
}

header{
background:linear-gradient(90deg,#567bd0,#2563eb);
color:white;
text-align:center;
padding:40px;
text-shadow:2px 2px 6px rgba(0,0,0,.4);
}

nav{
display:flex;
justify-content:center;
gap:30px;
background:#111827;
padding:18px;
position:sticky;
top:0;
}

nav a{
color:white;
text-decoration:none;
font-weight:600;
padding:10px 18px;
border-radius:30px;
transition:.4s;
}

nav a:hover{
background:#3b82f6;
color:#fff176;
transform:translateY(-3px);
}

.hero{
padding:35px 20px;
text-align:center;
}

.hero-content{
max-width:800px;
margin:auto;
background:rgba(255,255,255,.35);
backdrop-filter:blur(12px);
padding:40px;
border-radius:20px;
box-shadow:0 8px 20px rgba(0,0,0,.2);
```

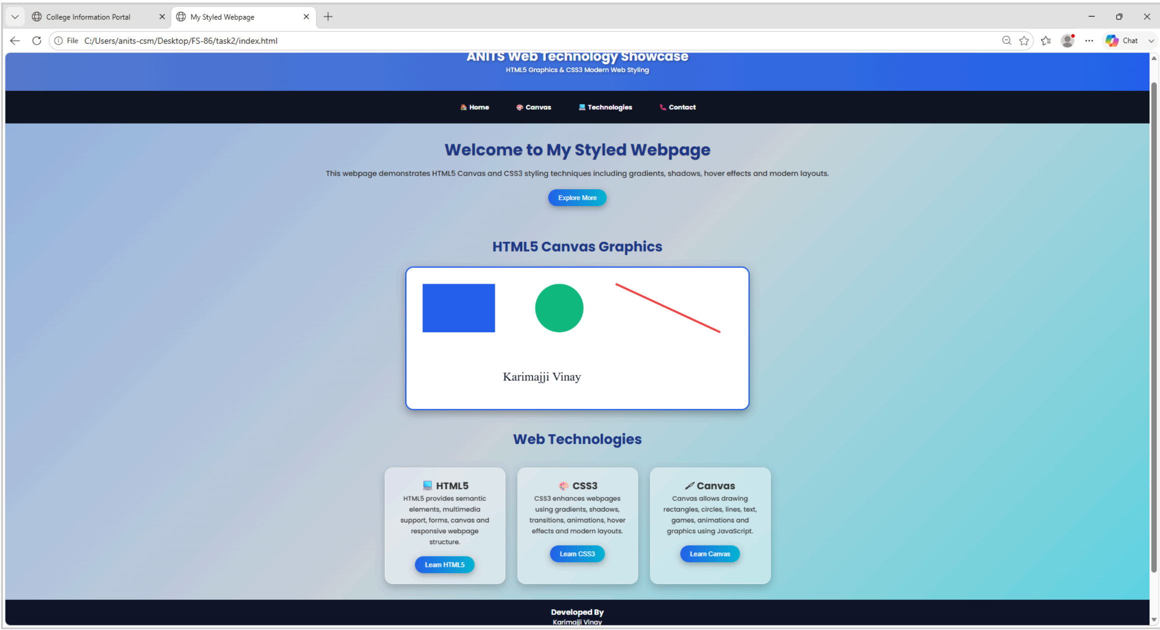
```
}

.hero h2{
font-size:40px;
margin-bottom:15px;
color:#1e3a8a;
text-shadow:1px 1px 2px #bbb;
}

.hero p{
font-size:18px;
margin-bottom:25px;
}

button{
padding:12px 25px;
border:none;
border-radius:30px;
background:linear-gradient(90deg,#2563eb,#06b6d4);
color:white;
font-size:16px;
cursor:pointer;
transition:.4s;
box-shadow:0 5px 15px rgba(0,0,0,.3);
}

button:hover{
transform:scale(1.08);
background:linear-gradient(90deg,#16a34a,#22c55e);
}

.section-title{
text-align:center;
font-size:34px;
margin:40px 0 25px;
color:#1e3a8a;
}

#canvas{
text-align:center;
}

canvas{
background:white;
border:4px solid #2563eb;
border-radius:20px;
box-shadow:0 8px 20px rgba(0,0,0,.3);
}

.card-container{
display:flex;
justify-content:center;
flex-wrap:wrap;
gap:30px;
padding:20px;
}

.card{
```

```
width:300px;
background:rgba(255,255,255,.45);
backdrop-filter:blur(12px);
padding:25px;
border-radius:20px;
border:1px solid rgba(255,255,255,.4);
box-shadow:0 8px 20px rgba(0,0,0,.2);
transition:.4s;
text-align:center;
}

.card:hover{
transform:translateY(-10px) scale(1.03);
}

.card h3{
font-size:26px;
margin-bottom:15px;
color:#2563eb;
text-shadow:1px 1px 2px #ddd;
}

.card p{
margin-bottom:20px;
line-height:1.7;
}

footer{
background:#0f172a;
color:white;
text-align:center;
padding:18px;
margin-top:20px;
}

@media(max-width:900px){
nav{
flex-wrap:wrap;
gap:10px;
}
.card-container{
flex-direction:column;
align-items:center;
}
.hero h2{
font-size:30px;
}
canvas{
width:95%;
height:auto;
}
}
```

8. TEST CASES AND EXECUTION DATA

Test Case	Input / Action	Expected Result	Actual Result	Status
TC-01	CSS Linking	Load index.html	External CSS styles applied correctly	**Pass**
TC-02	Flexbox Cards	Inspect card container	Cards displayed side-by-side with shadows	**Pass**
TC-03	Canvas Shapes	Observe canvas element	Shapes rendered at exact coordinates	**Pass**

9. OUTPUT



10. RESULT

Thus, the styled webpage with external CSS3 and interactive HTML5 Canvas graphics was successfully implemented and verified with expected visual rendering across all test cases.